

Uncovering evolutionarily remote and highly potent antimicrobial peptides with protein language models

Received: 15 November 2024

Accepted: 8 February 2026

Published online: 03 March 2026

 Check for updates

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Identifying evolutionarily remote antimicrobial peptides (AMPs) is crucial for discovering underexplored clinical candidates to combat antibiotic resistance. Existing experimental and computational methods are limited by their reliance on sequence identity to known AMPs, missing distant homologues. Here we introduce HMD-AMP, a protein language model-based approach for AMP discovery. HMD-AMP outperforms previous methods in identifying evolutionarily distant AMPs and enables the discovery of unknown and highly potent AMPs from metagenomic data. Applied to host and gut microorganism genomes of nine mammals, HMD-AMP revealed over 37 million predicted AMPs. Of 91 high-confidence sequences experimentally validated, 74 showed strong antibacterial activity and 48 were evolutionarily remote from known AMPs. Four of these AMPs exhibited broad-spectrum antibacterial activity at low effective concentrations and showed low toxicity, with the most potent peptide demonstrating therapeutic efficacy in a mouse model of peritoneal *Escherichia coli* infection. This study introduces an effective strategy to uncover AMPs.

The overuse of antibiotics has led to the emergence and spread of antimicrobial resistance, posing notable challenges to health care systems¹. Antimicrobial peptides (AMPs) constitute one of the most promising alternatives to antibiotics for the effective control of drug-resistant pathogens, as they have existed in nature for millions of years with almost no or limited resistance development. In recent decades, AMPs have gained substantial research interest. Scientific contributions of AMP discovery include a wide range of wet laboratory studies^{2–6}. While wet laboratory research is crucial for knowledge discoveries

in this domain, its limited generality makes it difficult to evaluate AMPs at scale. With the advancement of statistical and computational methodologies, machine learning frameworks are developed for AMP discovery^{7–17}.

Nevertheless, there are several major challenges with computational approaches. Known AMPs represent only a small portion of peptide sequence space, and short peptides can evolve rapidly, resulting in AMPs with highly diverse sequences¹⁸. Identifying evolutionarily remote AMPs is crucial for expanding our knowledge, understanding

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AMP structure and function and elucidating treatment mechanisms¹⁹. However, current AMP discovery methods struggled to detect remote and potent AMPs that differ from those already characterized. Moreover, most antimicrobial databases contain only positive (antimicrobial) sequences^{20,21}, making it difficult to generate high-quality negative samples. Finally, while natural AMPs—key components of animal innate immunity—are abundant and often potent and non-toxic^{22,23}, recent studies have focused mainly on microbiome-derived AMPs, potentially overlooking those from host genomes^{9,11–13}.

Here, we addressed the aforementioned challenges by proposing an end-to-end machine learning framework called protein language-powered Hierarchical Multi-task Deep forest (HMD-AMP) for AMP prediction and discovery. By curating a challenging training set and leveraging a fine-tuned protein language model (PLM), HMD-AMP captures evolutionary and functional patterns in peptide sequences. It outperforms previous methods by over 10% in Matthews correlation coefficient (MCC) for both AMP classification and functional type prediction, enabling high-resolution AMP discovery.

We applied HMD-AMP to 1,850 gut microorganism genomes from nine mammals, predicting about 35 million AMPs with less than 40% similarity to known AMPs. Focusing on the well-characterized pig gut microbiome, we identified 7,647 candidate AMPs from host and microbial genomes, synthesized 62 high-confidence peptides and found 52 with strong antibacterial activity. Further validation in three other mammals yielded 22 effective AMPs out of 29 tested. Overall, these AMPs display diverse structures and low sequence similarity to known AMPs. We found eight AMPs, including four evolutionarily remote and one sharing only 9% identity with any known AMP, that showed broad-spectrum, low-toxicity activity, comparable to polymyxin B and vancomycin (Fig. 1a). Finally, a mouse model infected with *Escherichia Coli* in the peritoneum confirmed that the one evolutionarily remote AMP with the strongest antibacterial effect retains antibacterial activity in vivo (Fig. 1b). Our study highlights the power of PLMs for discovering diverse, potent and evolutionarily remote AMPs.

Results

Using PLMs for evolutionarily remote AMP identification

PLMs can encode the evolutionary features of proteins and aid in detecting remote homology^{24,25}. Inspired by that, we built a pipeline using PLMs for evolutionarily remote AMP discovery from metagenomes (Fig. 1a). Data screening in the pipeline comprised length filtering, redundancy removal, AMP prediction, metaproteome filtering and ranking by prediction score. Subsequently, promising candidates were synthesized and subjected to in vitro and in vivo validation assays (Fig. 1b).

Within the pipeline, HMD-AMP, a PLM-based deep learning framework, is developed as the core module for AMP identification. It takes raw amino acid (AA) sequences as input and obtains the embeddings generated by PLM (ESM-2 (ref. 26)) first. Considering the PLM was trained mainly on proteins longer than 100 AAs (Supplementary Fig. 1) but AMPs are mostly between 5 and 50 AAs, we fine-tuned the PLM (Supplementary Table 1) on short-peptide data to assist AMP prediction tasks. Then, the embeddings are fed into the prediction module²⁷. The model first classifies sequences as AMPs or non-AMPs and then further predicts their biological activities (Fig. 1c).

HMD-AMP can recognize AMPs from remote organisms and those with potent antibacterial activity

Building on this framework, we evaluated HMD-AMP and demonstrated its feasibility in uncovering highly potent AMPs and AMPs from unknown organisms. A total of six representative AMP prediction methods that include both traditional machine learning models and deep learning models, namely, AMAP²⁸, AmPEP²⁹, i-AMP2L³⁰, AMP Scanner Vr.2³¹, ATT⁹ and BERT⁹, were included for comparison. They were re-trained on the same training data as ours to ensure a fair

performance assessment. We first analysed our performance using the fivefold cross-validation strategy on the training set. Detailed results are shown in Fig. 2a and presented in Supplementary Table 2. HMD-AMP outperformed all other methods across all evaluation metrics, except for recall, where its performance was comparable to that of AMP Scanner. For accuracy, precision and F1-score, we surpassed the second-best method by approximately 0.02. Since homology partition is not applied in cross-validation, all models performed highly and performance differences were not pronounced.

We then sought more challenging tests and evaluated our model using the leave-one-out experiment across different kingdoms. We first obtained the source organism information of our training data by searching for DBAASP and Uniprot³² and then reconstructed the phylogeny of sequences with organism information (Fig. 2b). The results show that the training data are derived from various organisms and have diverse evolutionary branches, constructing a species-comprehensive training resource. We grouped the training data by source organism: animalia, plantae, bacteria and others. In each iteration, one group was used for testing and the rest for training. As shown in Fig. 2c and presented in Supplementary Tables 3–5, HMD-AMP performed the best in all groups. For animalia and plantae, all deep learning-based methods showed decent performance. However, for the bacteria group, HMD-AMP outperformed other models on the F1-score by a substantial margin of at least 0.17. This is probably because bacterial sequences had much lower similarity to the training data (mostly <50%), compared with animalia and plantae (up to 90%; Supplementary Fig. 4). Overall, HMD-AMP demonstrated robustness across different kingdoms and can precisely identify AMPs and non-AMPs from various organisms.

We also assessed whether HMD-AMP's prediction scores reflect AMP potency by analysing minimum inhibitory concentration (MIC)-based experimental data from DBAASP (only μM units; MIC values >256 were capped at 256). Figure 2d shows the relationship between HMD-AMP's confidence score and MIC. AMPs with a confidence score ≥ 0.9 generally have lower MIC values, indicating stronger antimicrobial activity. Notably, for the highly potent AMPs with MIC values below 4 μM , 93.3% have confidence scores exceeding 0.99, whereas for AMPs with MIC values >64 μM , the proportion drops to approximately 20%. Consequently, high HMD-AMP scores reliably indicate potent AMPs. The results support that HMD-AMP can precisely identify AMPs from various organisms and distinguish the potency of AMPs.

HMD-AMP is robust in identifying AMPs sequentially and structurally distant from the training data

To further assess HMD-AMP's generalization and remote AMP detection, we assembled an independent test set from sources orthogonal to the training data, benchmarking performance by length, sequence and structure. Examining performance by AMP length (Fig. 3a), HMD-AMP was superior to other models across all groups. All methods showed decreased performance for AMPs shorter than 15 AAs. For longer sequences, HMD-AMP maintained an F1-score above 0.88, while others varied. Deep learning-based models predicted AMPs longer than 45 AAs more accurately than those between 30–45, whereas traditional machine learning models showed the opposite trend, probably owing to compromised sequence integrity in their input features, limiting long-range contextual information capture.

Recognizing the importance of detecting AMPs with low sequence and structure similarity to known examples, we conducted a sequence-similarity cut-off test (Fig. 3b). We performed seven tests, each excluding test data with a sequence similarity above cut-off values to any training sample. As the threshold decreased, all models showed reduced performance. For sequences with less than 40% identity to training data, all models except HMD-AMP had F1-scores below 0.65. By contrast, HMD-AMP experienced only a marginal decline, maintaining at least a 0.13 advantage in F1-score over others and a drop of no more than 0.1 compared with the 90% similarity cut-off.

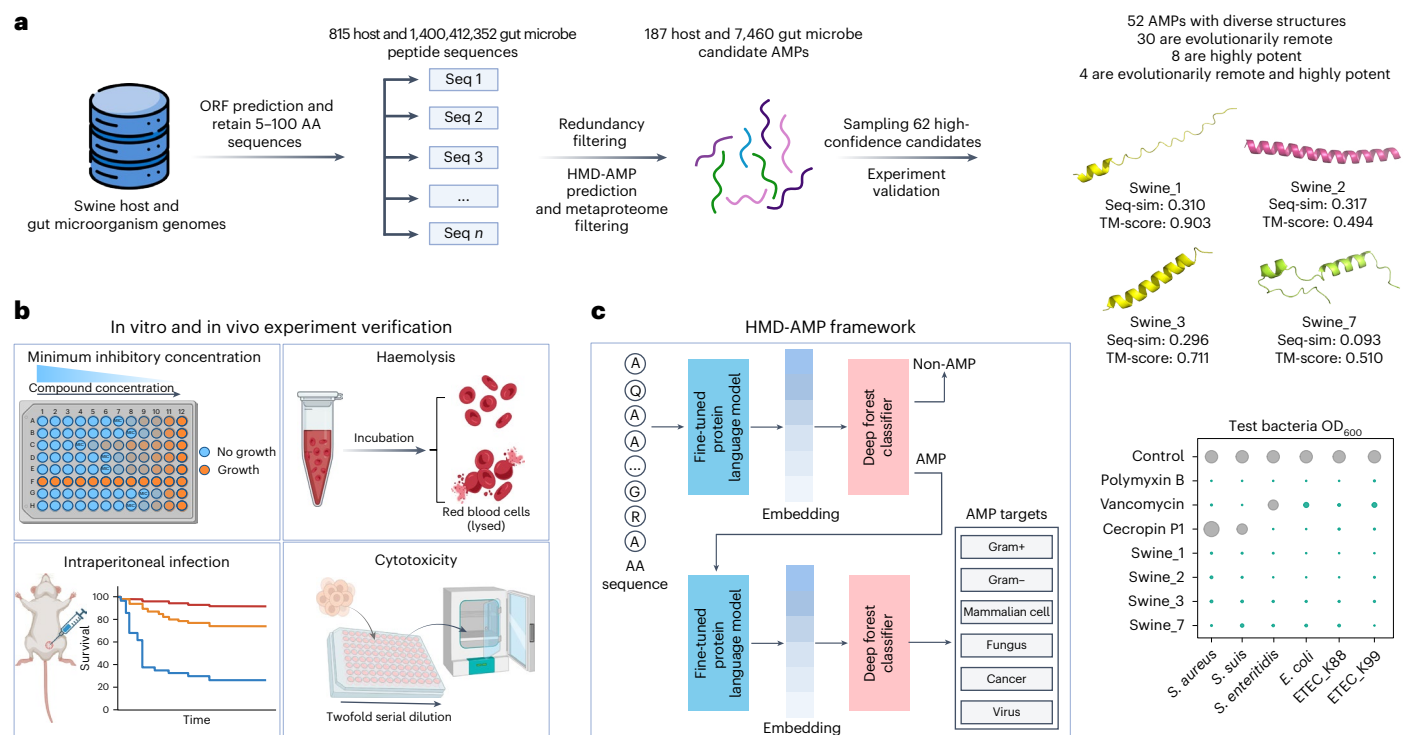


Fig. 1 | The workflow for discovering evolutionarily remote and highly potent AMPs. **a**, Our pipeline starts with collecting metagenomic data to predict ORFs. Representative ORFs with lengths between 5 and 100 AAs then go through the AMP predictor, HMD-AMP, for unknown AMP mining. After that, metaproteome filtering is applied to obtain candidate AMPs, and some candidates with high predicted scores are sampled for chemical synthesis and wet laboratory-experiment validation. With the help of a PLM, the pipeline can discover AMPs with diverse structures and highly potent antibacterial activity. Here, we visualized tertiary structures of four discovered evolutionarily remote and highly

potent AMPs and listed their OD₆₀₀ values against six bacteria at a concentration of 60 μ M (Swine_7 is host derived, and seq-sim and TM-score indicate the sequence similarity and TM-score between discovered AMPs and the sequentially and structurally closest training AMPs). **b**, In vitro and in vivo experiments to verify the AMP efficacy. **c**, The HMD-AMP framework. It takes raw AA sequences as input and uses a PLM as a feature extractor; subsequently, a hierarchical deep forest classifier is applied to predict AMP and its targets. The detailed framework is shown in Supplementary Fig. 2. Diagrams in **a** and **b** created in BioRender; Group, A. <https://BioRender.com/70nwfyg> (2025).

We subsequently conducted the structure-similarity cut-off test, using five TM-score³³ thresholds from 0.9 to 0.5 (Fig. 3c). HMD-AMP remained stable across all cut-off values. When predicting AMPs not similar to training data (TM-score under 0.5), the performance gap between HMD-AMP and other models was the largest, achieving a 0.12 advantage in the F1-score.

Then, we delved into the performance of different models on the 40% sequence-similarity cut-off set. As presented in Supplementary Table 6 and shown in Supplementary Fig. 5, HMD-AMP demonstrated considerable improvements across all evaluation metrics in identifying these remote AMPs. Especially for precision and MCC, our method outperformed other models by at least 0.1. A detailed comparison with the second-best model, BERT, on both the 40% sequence similarity and 0.5 TM-score cut-off tests (Supplementary Table 7) showed that HMD-AMP outperformed BERT by at least 0.1 on precision, recall and F1-score.

We calculated the highest structural similarity (TM-score) between AMPs in the 40% sequence-similarity cut-off test set and those in the training set (Fig. 3d). Results showed that even AMPs with a very low sequence similarity (below 10%) could have high structural similarity, with 79.1% of data points having TM-scores above 0.5. HMD-AMP achieved high accuracy for AMPs with both very high and very low TM-scores (Supplementary Table 8). The predicted structures had relatively high predicted local distance difference test (pLDDT) scores, averaging above 70 (Supplementary Fig. 6). A recent study³⁴ found that AlphaFold2 often gives high pLDDT scores to short peptides even when predictions are inaccurate, which may explain the high accuracy at TM-score extremes. Subsequent analysis showed that HMD-AMP

uniquely detected 18 AMPs missed by other models (Fig. 3e). These AMPs had different motifs from training AMPs and distinct AA composition distributions from other AMPs in the 40% sequence-similarity cut-off test set (Supplementary Fig. 7).

Finally, we conducted an ablation study to assess the contribution of each module in HMD-AMP. Visualization of embeddings showed that ESM-2, especially after fine-tuning, more clearly separated AMPs from non-AMPs compared with one-hot encoding (Supplementary Fig. 8). We tested various model configurations by removing or replacing the fine-tuning step, the ESM-2 model and the deep forest module. Results confirmed the strong impact of the PLM, and other modules also contributed to performance improvements (Supplementary Fig. 9 and Supplementary Table 9). In addition, balancing the training set greatly enhanced model prediction (Supplementary Fig. 10 and Supplementary Table 10). In summary, the PLM substantially boosts HMD-AMP's generalization, enabling effective identification of evolutionarily distant and structurally diverse AMPs.

HMD-AMP exhibits substantial improvements in predicting AMP targets

Understanding AMP functional types is crucial for elucidating their mechanisms. Accurate annotation tools improve the reliability of in silico screening. Here, we compared HMD-AMP with three existing models^{28,30,35} on six functional types: Gram-positive, Gram-negative, mammalian cell, cancer, virus and fungus. Most labels had imbalanced data, particularly cancer and virus, where negatives outnumbered positives by about five to one (Fig. 3f). HMD-AMP improved MCC by at

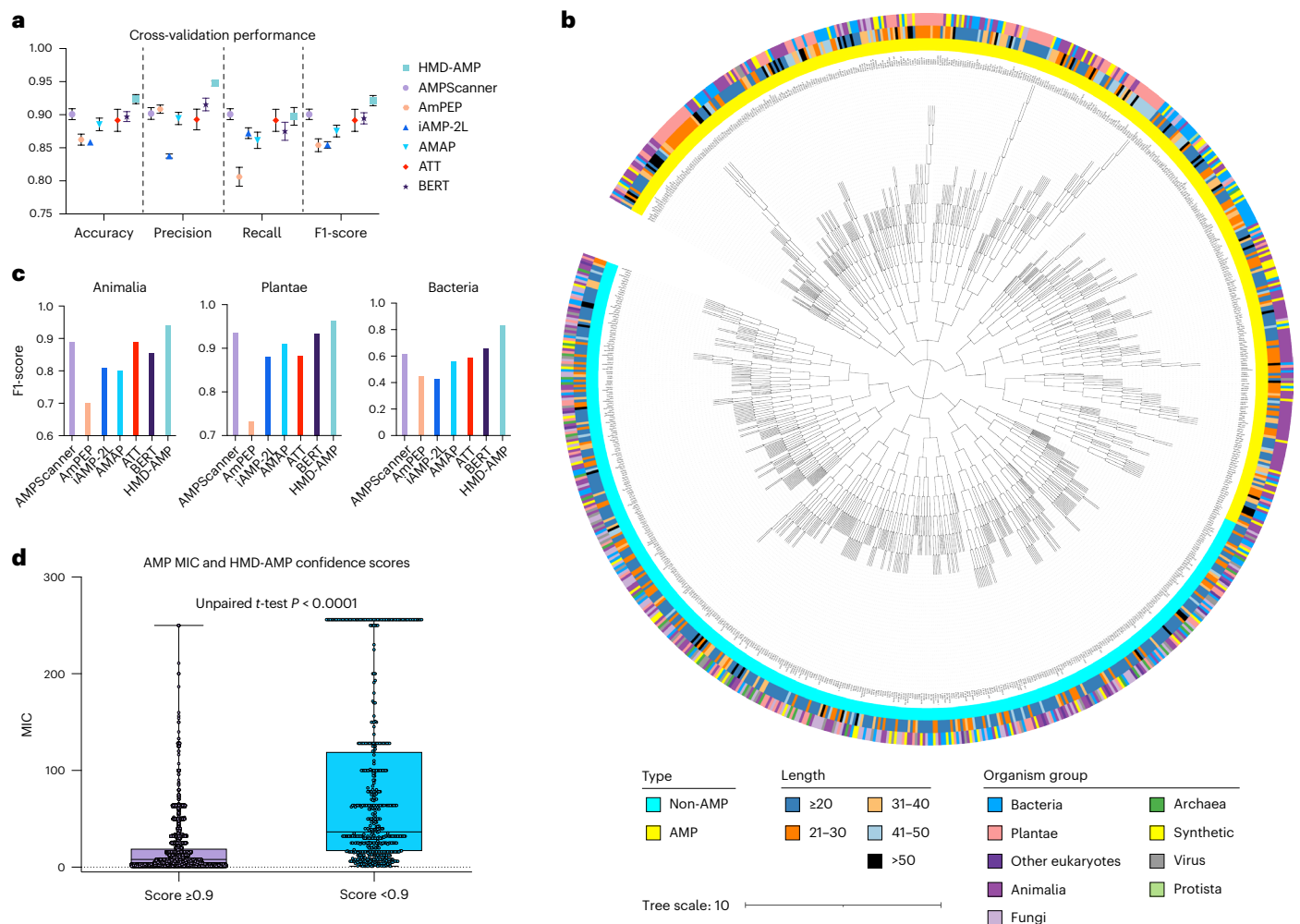


Fig. 2 | HMD-AMP outperforms other models on cross-validation and cross-kingdom test. a, Point plot of performance of HMD-AMP and other models on fivefold cross-validation. The sample size of each point is 5. Data are presented as mean $\pm$ s.d. A two-sided paired *t*-test shows the overall performance of HMD-AMP is significantly different from all models (AmPEP, $P = 0.0093$; iAMP-2L, $P = 0.0302$; AMAP, $P = 0.0015$; BERT, $P = 0.0007$) except AMP Scanner ($P = 0.1196$) and ATT ($P = 0.538$); these two models show comparable performance to our method because cross-validation lacks homology partition. **b**, Cladogram of species tree of samplings in training data. The annotation rings from inner to outer are: (1) data type, (2) length of AMPs and (3) organism group of peptides. The phylogenetic tree of all the training AMPs is shown in Supplementary Fig. 3.

c, Performance of HMD-AMP and other models on cross-kingdom tests. Here, we constructed three test sets that contain data just from animalia, plantae and bacteria, respectively. The sample size is 3 and we displayed the mean value. A two-sided paired *t*-test shows the performance of HMD-AMP is significantly different from all models on all kingdoms ($P < 0.05$). **d**, The box plot of confidence scores of AMPs and their corresponding lowest MIC on record. AMPs with a confidence score ≥ 0.9 ($n = 6,334$) have a lower MIC than those with a confidence score < 0.9 ($n = 537$) (two-sided unpaired *t*-test $P < 0.0001$). Box plots are defined as: centre line, median (50th percentile); bounds of box, 25th–75th percentiles; whiskers, data minima and maxima.

least 0.05 across all types and by 0.1 for Gram-positive, Gram-negative and mammalian cell targets (Fig. 3g, Supplementary Fig. 11 and Supplementary Tables 11–16).

We observed a striking similarity in the AA composition between AMPs targeting Gram-positive and Gram-negative bacteria, while anti-mammalian cell AMPs displayed a higher proportion of Ala, Leu and Lys and fewer Asp, Glu, Arg and Tyr (Fig. 3h). Although AMAP, iAMP-2L and AMPfun all included AA composition as input features that help distinguish function types, the performance of these models remains much lower than HMD-AMP.

Evolutionarily remote candidate AMPs are uncovered from metagenomic data

Previous studies mainly focused on screening from short peptides^{10,16,17} or common microorganism data^{9,12,13}. As animals naturally produce AMPs as part of their innate immunity, we hypothesized that host data are great resources for potent and non-toxic AMPs.

First, to assess HMD-AMP's extrapolation ability for discovering remote AMPs, we selected 1,850 genomes from gut microorganisms of unknown species, spanning nine mammalian hosts across different genera (Supplementary Table 17). The genomes yielded 206,356,600 peptides, of which HMD-AMP predicted 37,461,558 AMPs with scores above 0.6. Most of these had less than 40% sequence identity to training AMPs (Supplementary Fig. 12).

Then, given the comprehensive genome data available for the swine gut microbiome and host^{36,37}, and the urgent need for new therapeutics in swine production owing to rising antimicrobial resistance³⁸, we focused on swine and its gut microorganisms to identify AMPs as viable antibacterial alternatives. Specifically, 14,026 microbial genomes from the swine gut and 1 host genome were collected (Fig. 4a). After open reading frame (ORF) prediction, we obtained 815 host peptides and 1,400,412,352 gut microorganism peptides (5–100 AAs). To further manage data size, we applied redundancy filtering at 95% sequence identity and coverage, resulting in 815 host-derived and 584,645,352

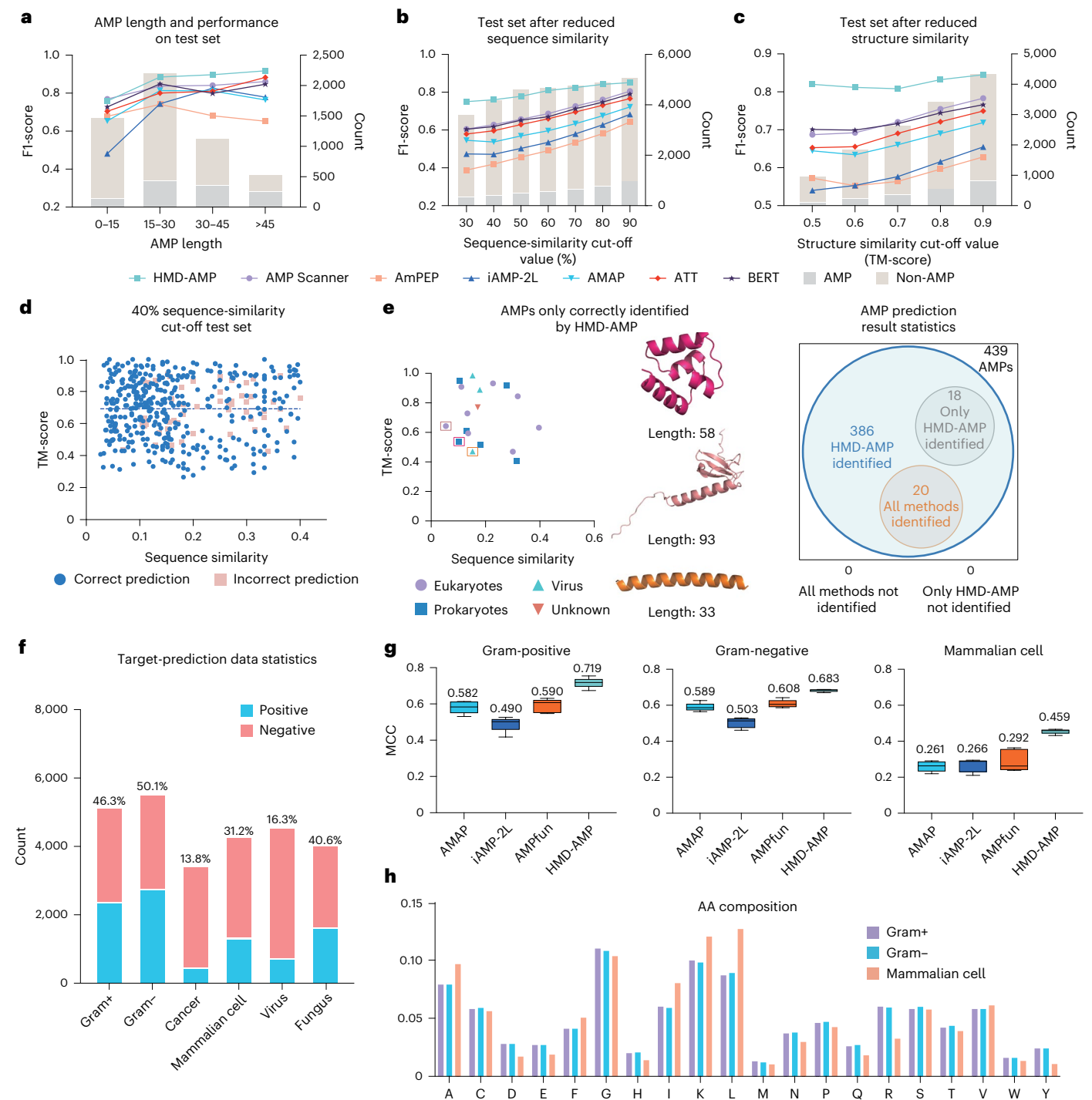


Fig. 3 | HMD-AMP shows robust performance on homology-partitioned

independent test and AMP target annotation tasks. a, The F1-score of models on AMPs of various lengths, along with the counts of AMPs and non-AMPs in each length range. Most AMPs are below 45 AAs in length, and HMD-AMP showed stable performance across different lengths. **b**, The performance of models on the test set at different sequence similarity cut-off values. We tested seven cut-off values, and HMD-AMP consistently outperformed other models in terms of F1-score. **c**, The performance of models on the test set at various structure-similarity (TM-score) cut-off values. We set five cut-off values, and HMD-AMP performed the best in all tests. **d**, The sequence similarity and TM-score of AMPs in the 40% sequence-similarity test set against the sequentially and structurally closest training AMPs. Correct and incorrect prediction means accurately or inaccurately predicted by HMD-AMP. **e**, Prediction result statistics of AMPs in the 40% sequence-similarity

cut-off set. There were 18 AMPs accurately predicted by HMD-AMP but not identified by all the other models. We show their source organism as well as sequence similarity and TM-score to their closest AMPs in the training data. In total, three shown tertiary structures correspond to the AMPs in the scatter plot with boxes of the same colour. **f**, The proportion of positive and negative data for different labels in the AMP target-prediction task. Data imbalances are prevalent. **g**, Box plot of the MCC performance of models in the AMP target-prediction task. We used fivefold cross-validation. The sample size for each box is 5. The two-sided paired *t*-test shows the performance of HMD-AMP is significantly different from all models across all functional types ($P < 0.05$). Box plots are defined as: centre line, median (50th percentile); bounds of box, 25th–75th percentiles; whiskers, data minima and maxima. **h**, AA composition of collected AMPs targeting Gram-positive bacteria, Gram-negative bacteria and mammalian cells.

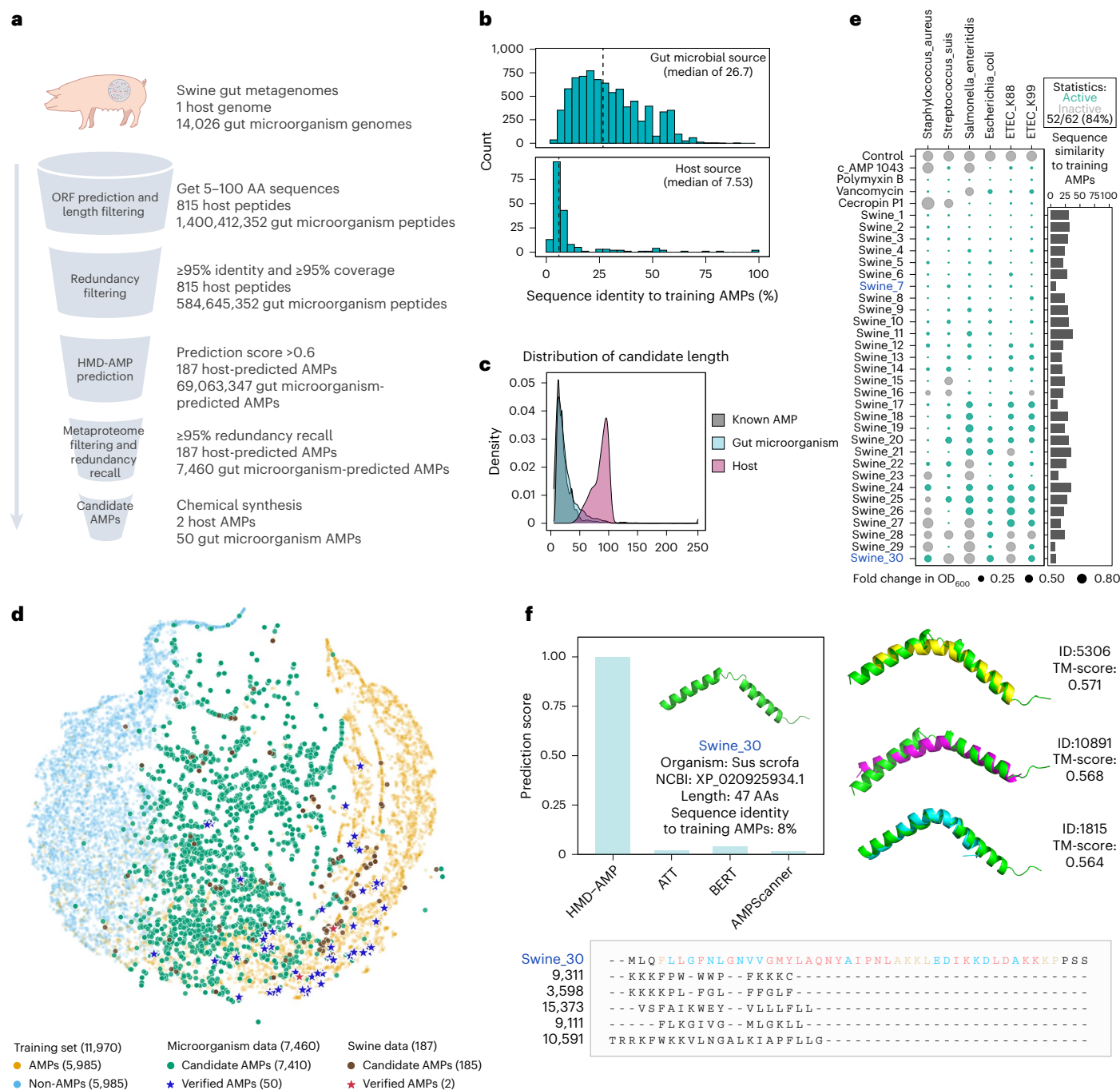


Fig. 4 | The discovery of highly diverse and potent AMPs from swine metagenomic data by HMD-AMP. a, Schematic representation of the application workflow. A total of 52 sequences were validated as potent AMPs. **b**, Sequence identity of discovered candidate AMPs to their closest AMPs in the training set. **c**, Length distribution of our discovered candidate AMPs from gut microorganism and host and known AMPs. **d**, Visualization of the embedding space that includes our training data and candidate AMPs discovered from swine microorganism and host data with t-SNE. Training AMPs and non-AMPs are largely separated, and candidates do not overlap much with the training data. **e**, Among 52 newly discovered AMPs, 30 are sequentially remote to the training data (sequence identity lower than 40%). Here, we show the bacterial inhibition spectra and level of these 30 AMPs against six swine-associated pathogens,

‘active’ (cyan colour) indicates that the AMP inhibits the growth of bacteria for a more than 50% reduction in OD₆₀₀ at a concentration of 60 μM ($P < 0.05$ in Dunnett’s test, two-sided, Swine_7 and 30 are from the host and others are from the gut microorganism). **f**, The prediction score, properties and tertiary structure visualization of Swine_30, which is correctly identified by HMD-AMP but failed by all other deep learning methods. On the right is the structure alignment of three AMPs in the training set that have the highest TM-score to Swine_30. At the bottom, a multiple sequence alignment of Swine_30 and its five most similar AMPs from the training set is shown. The conserved domain identified in Swine_30 is DUF4535 (Pfam: pfam15054; E-value of 1.06×10^{-9}). The alignment is coloured by residue conservation: red indicates high conservation, blue indicates low conservation and yellow indicates the aligned part.

gut microorganism-derived peptides. We sampled sequence pairs with >95% identity. For each pair, where one was retained and the other excluded during redundancy filtering, we found their HMD-AMP prediction scores were highly similar (Supplementary Fig. 13), confirming

that redundancy filtering preserved representative sequences while reducing data volume.

After redundancy filtering, HMD-AMP predicted 187 host-derived and 69,063,347 microorganism-derived AMPs. As gut microorganism

small ORFs (sORFs) were predicted in silico and not all may be expressed, we cross-referenced these AMPs with swine metaproteomic data, identifying 1,846 as candidate AMPs. Then, to gain comprehensive phylogenetic insights, we retrieved all sequences removed during redundancy filtering that shared over 95% identity and coverage with the 1,846 candidates, finally resulting in a set of 7,460 sequences. Through bacterial origins analysis, we found that most AMP candidates originated from the dominant swine gut phyla, Firmicutes and Bacteroidetes (Supplementary Fig. 14a), which together comprise 52% of our swine gut microbial genomes³⁹. Nearly 95% of Methanobacteriota genomes contained candidate AMPs, highlighting this phylum as a previously underappreciated source for AMP discovery. In addition, certain genomes, such as *Streptococcus alactolyticus*, harboured three or more AMP candidates (Supplementary Fig. 14b), suggesting specific taxa may be particularly rich in these peptides.

To investigate the sequence remoteness of candidate AMPs, we analysed sequence identity and length distribution. About half of gut microorganism candidates shared less than 25% identity with the closest training AMPs, while most host candidates had less than 10% identity (Fig. 4b). Candidates also showed low similarity to non-AMPs in the training set (Supplementary Fig. 15). Besides, we found that HMD-AMP prediction scores remained consistent across different sequence identity levels, indicating that predictions are not based on sequence similarity (Supplementary Fig. 16). Analysis of candidate AMP lengths showed gut microorganism sequences matched the training set distribution, while host candidates were notably longer (Fig. 4c and Supplementary Fig. 17). In addition, host candidates exhibited lower hydrophobicity (Supplementary Fig. 18), which could reduce interaction with neutral host cell membranes and promote selective antibacterial activity. Motif analysis further revealed wide variation among microorganism candidates, host candidates and training AMPs (Supplementary Fig. 19).

Then we analysed the remoteness of our discovered candidates from the perspective of model encoding and visualized the embedding space of our training data and candidate AMPs using t-SNE⁴⁰ (Fig. 4d). The results revealed known AMPs largely separated from non-AMPs, and embeddings of most candidates were different from training AMPs, suggesting HMD-AMP generalize to unseen embedding spaces. From both sequence and embedding, HMD-AMP extrapolates from its training regime to unknown knowledge.

Experimental validation confirms high proportion of evolutionarily remote and potent AMPs

We selected 2 host and 60 microorganism candidate sequences for chemical synthesis and antibacterial activity validation. The host-derived sequences had the highest HMD-AMP prediction scores. For microorganism candidates, we chose sequences predicted as AMPs by at least two out of AMP Scanner, ATT and BERT and with HMD-AMP scores above 0.9. For antibacterial activity testing, we chose two Gram-positive and four Gram-negative priority pathogens (including *Staphylococcus aureus*, *Salmonella enteritidis*, *Streptococcus suis*, *E. coli* and enterotoxigenic *E. coli* (ETEC)) that pose global and regional threats to swine health⁴¹. The function of AMPs was tested at a 60- $\mu\text{mol l}^{-1}$ level in Mueller–Hinton broth (MHB) medium. For antibacterial activity comparison, we included two peptide drugs, polymyxin B and vancomycin, a broad-spectrum swine gut AMP, cecropin P1⁴², and an human gut potent AMP, c AMP1043⁹.

All peptides were successfully synthesized, and 52 (2 from the host and 50 from the microorganism) showed potent antibacterial activity (could reduce the optical density at 600 nm (OD_{600}) index of at least one bacteria by more than 50% at a concentration of 60 μM), reflecting an 84% potent AMP positive rate (Fig. 4e and Supplementary Fig. 20). The phylogenetic tree of 52 AMPs displayed the diverse evolutionary branches (Supplementary Fig. 21). Among them, 27 AMPs showed potency to all pathogens, with some matching the antibacterial levels of polymyxin B and vancomycin. In total, 51 AMPs inhibited Gram-negative bacteria and

46 AMPs were effective against Gram-positive bacteria. The prediction performance (Supplementary Table 18) of HMD-AMP on the functional types of 62 synthesized peptides is consistent with the performance shown in Fig. 3h. Here, 57.7% (30/52) validated AMPs had less than 40% sequence identity to their closest training data; the median was 32.5%. A total of four AMPs had even less than 10% sequence identity, indicating their divergence from existing knowledge. We also chose three sequences predicted as non-AMPs by HMD-AMP as negative samples to synthesize and did not find notable antibacterial effects (Supplementary Fig. 20).

In analysing the prediction results of HMD-AMP and the other three models on these 52 peptides (Supplementary Table 19), the false negatives produced by the three models are at least doubled when predicting evolutionarily remote AMPs compared with AMPs that have higher sequence similarity to training data. In addition, they struggled to recognize the two host-derived AMPs. Swine_7 was missed by ATT and BERT. Swine_30 was only correctly identified by HMD-AMP (Fig. 4f). Despite being trained on the same data, the three models assigned Swine_30 very low scores. A possible explanation is Swine_30's exceptional remoteness, with maximum sequence identity and TM-score to training data at just 8% and 0.571, respectively. The top 3 training AMPs that had the highest TM-score were all structurally different from Swine_30 (each has one α -helix), and the sequence alignment of Swine_30 to the top 5 sequentially closest training AMPs showed notable variations. Conserved domain analysis⁴³ revealed Swine_30 belongs to the domain of unknown function (pfam15054: DUF4535, E-value of 1.06×10^{-9}), suggesting that this family may harbour antibacterial sequences. Conserved residue analysis further highlighted the uniqueness of Swine_30 compared with known AMPs, which probably contributed to it being missed by other models. By contrast, HMD-AMP was able to capture evolutionary information beyond annotated AMP data and correctly identify such remote AMPs.

To further test the generalizability of HMD-AMP, we validated predicted AMPs (Supplementary Table 17) from gut microorganisms of three other hosts. They are *Bos taurus* (cattle), a domesticated livestock species; *Homo sapiens* (Hadza hunter-gatherers), whose microbiome contains evolutionarily remote taxa relative to urban populations⁴⁴; and *Papio cynocephalus* (yellow baboon), a wild primate known to harbour undescribed microbial lineages⁴⁵. We experimentally assessed 29 top-scoring peptides against ETEC K88 and *S. aureus* American Type Culture Collection (ATCC) 6538. Of these, 22 of 29 exhibited potent antimicrobial activity, and 18 have less than 40% sequence identity to known AMPs (Supplementary Fig. 22).

Among 74 validated AMPs, 48 are evolutionarily remote (Fig. 5a). Structure predictions classified remote AMPs into six categories on the basis of the presence or absence of α -helices and β -sheets, including α -helical, extended, β -sheet and mixed types. Despite their low sequence similarity to known AMPs, most still adopt canonical AMP structural folds (Fig. 5b), suggesting convergent evolution towards effective antimicrobial architectures. For example, Bos_8, a *Bos taurus* gut microbiota-derived AMP, is an α -helical peptide. Its secondary structure and helical wheel (Fig. 5c) show a classic amphipathic topology, with hydrophobic residues clustering on one side of the helix and polar or charged residues on the other. This spatial arrangement facilitates membrane insertion and disruption, which is a hallmark mechanism of α -helical AMPs such as Magainin 2. The results suggest evolutionarily remote peptides can retain structural and mechanistic characteristics of established AMPs. Beyond structural conservation, we used STREME⁴⁶ to identify sequence motifs shared between remote and known AMPs. Four conserved motifs were found, with two especially prevalent, present in 39.6% and 25% of remote AMPs, respectively. The most common motif is characterized by alternating positively charged (K and R) and hydrophobic (L) residues, which are typical features of amphipathic membrane-active peptides (Fig. 5d). The second most common motif is cysteine-rich (C), probably facilitating disulfide bond formation and structural stabilization, akin to defensins such as θ -defensins (Supplementary Fig. 23).

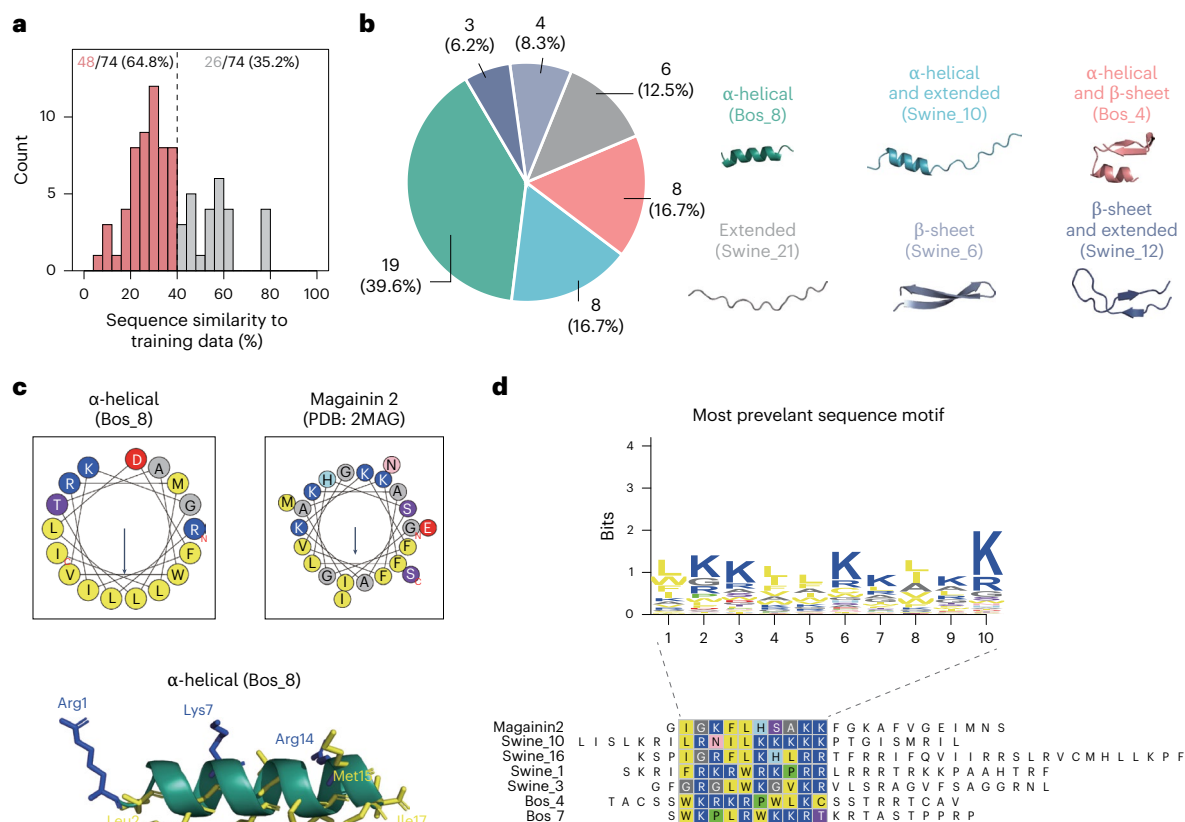


Fig. 5 | Structure and sequence features of evolutionarily remote AMPs associated with antimicrobial function. **a**, Sequence identity of 74 validated AMPs relative to the training dataset. In total, 48 AMPs exhibit less than 40% identity and are evolutionarily remote. **b**, Distribution of the 48 remote AMPs according to their structural classes (left). Representative structural examples for each class (right). **c**, Helical wheel plots of a representative remote AMP and a reference AMP, magainin 2 (top). Hydrophilic residues (D, N, K, S, T and R)

cluster on one side and hydrophobic residues (F, I, L, M, V and W) on the other. Blue, positively charged; red, negatively charged; yellow, hydrophobic; green, polar uncharged. Arrows denote Eisenberg hydrophobic moments. Bottom, visualization of the tertiary structure of the representative AMP. **d**, The most prevalent sequence motif identified by STREME⁴⁶ in both remote AMPs and existing AMPs. We showed six remote AMPs and one reference AMP (Magainin 2) that contained this motif.

These functional analyses show that, although remote AMPs have low sequence similarity to known AMPs, they still retain conserved structural and sequence features essential for antimicrobial activity. This highlights HMD-AMP's ability to detect remote homologues and uncover biologically meaningful AMPs with conserved mechanistic properties.

The most potent AMP has low toxicity and is effective against peritoneal infection in a mouse model

From 52 synthesized swine-derived AMPs, 14 with strong inhibition against all six pathogens were selected for MIC testing against the four most prevalent swine pathogens. The antibacterial capacity curves are shown in Supplementary Fig. 24. For Gram-positive species, ten AMPs presented a MIC <16 μM, and the minimal MIC reached 2 μM in *S. suis*. A similar trend could also be observed in *S. aureus*. For Gram-negative species, including *S. enteritidis* and ETEC K88, seven AMPs exhibited a MIC less than 16 μM, with the minimum value being only 1 μM. In our assay, eight AMPs demonstrated exceptional antibacterial activity against all pathogens, with MIC values comparable to vancomycin and polymyxin B (Fig. 6a and Supplementary Table 20). Five AMPs achieved MIC values below 16 μM against all four bacterial species. Notably, among eight highly potent AMPs, Swine_1, Swine_2, Swine_3 and Swine_7 have low sequence similarity to known AMPs.

We assessed the potential toxicity of the most potent evolutionarily remote AMPs: host-derived Swine_7 and microorganism-derived Swine_2 and Swine_3. Haemolysis assays on swine red blood cells showed

no haemolytic effects at any concentration (Fig. 6b). In cytotoxicity tests on HT-29 cells, Swine_3 and Swine_7 only exhibited cytotoxicity at 128 μmol l⁻¹, well above their MIC values, while Swine_2 showed only slight cytotoxicity even at high concentrations. Five additional potent AMPs (Supplementary Fig. 25) also showed no notable toxicity. These results support the notion that host-derived AMPs are probably non-toxic.

To investigate the bactericidal mechanism, we used scanning electron microscopy (SEM) to monitor the morphological change of ETEC K88 cells after 3 h of treatment with three AMPs. As shown in Fig. 6c, Swine_2 and Swine_7 had a strong destructive effect on the cellular structure, with a large amount of cell debris observed, whereas cells treated with Swine_3 showed depression in the microscopic view.

Finally, we evaluated the efficacy of discovered AMPs in vivo. A mouse model of peritoneal *E. coli* infection was used. One of the most potent AMPs against ETEC K88, Swine_2, was selected for testing. Gentamicin was used as a positive control. As shown in Fig. 6d, Swine_2 maintained antibacterial activity in vivo, increasing the survival rate of mice to 30%. Swine_2 deserves further investigation, as it is a prototype peptide without modification. It could potentially be optimized to enhance the efficacy in vivo⁴⁷.

Discussion

PLMs enrich sequence representations and have improved generalization in various protein function prediction tasks^{48,49}. This study shows PLMs enable HMD-AMP to detect remote AMPs and enhance

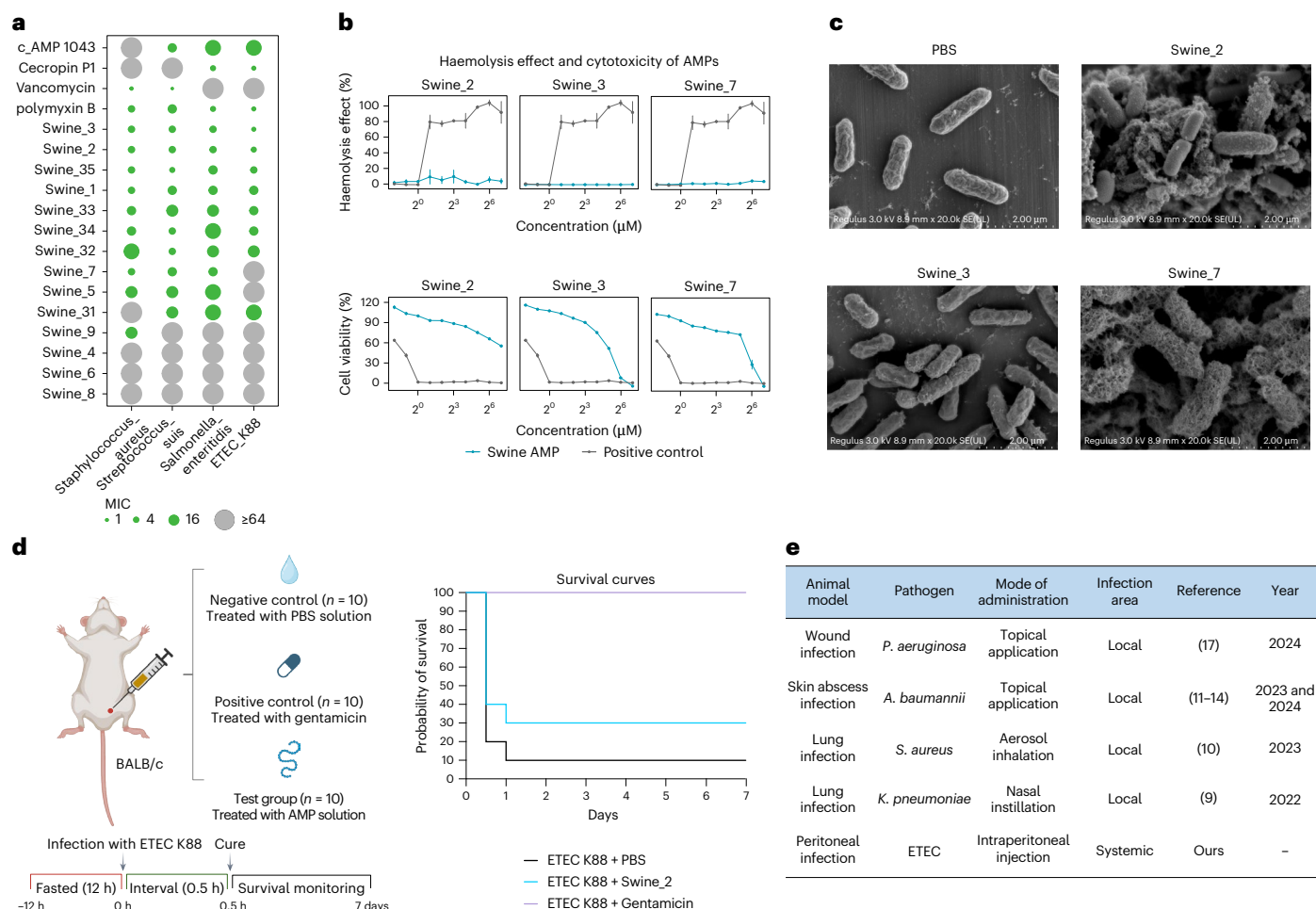


Fig. 6 | Characterization of the most potent AMPs discovered from the swine metagenomes. a, MICs of 14 selected AMPs against 4 prevalent swine pathogens. **b**, Results of haemolysis effect test and cytotoxicity test of the two most potent microorganism-derived AMPs and one host-derived AMP. A total of ten concentration gradients (128, 64, 32, 16, 8, 4, 2, 1, 0.5 and 0.25 $\mu\text{mol l}^{-1}$) were tested, and triton-X100 was used as the positive control for both tests. Data are presented as mean $\pm$ s.e.m. from three independent biological replicates ($n = 3$).

c, The s.e.m. result of ETEC K88 cells after 3 h of treatment with the two most potent microorganism-derived AMPs and one host-derived AMP; all images show decreasing cell integrity. **d**, In vivo experiment procedure; we built a mouse model subjected to peritoneal *E. coli* infection. The image also shows the survival curve of Swine_2. **e**, Differences between our designed in vivo experiment and those of previous AMP discovery methods. Diagram in **d** created in BioRender; Group, A. <https://BioRender.com/70nwfyg> (2025).

its generalization ability. In addition, our curated training data further improved performance. For the negative training set, we selected UniProt peptides annotated as ‘cytoplasm’ and excluded those with antimicrobial annotations. However, lack of annotation does not guarantee lack of antimicrobial activity, so some false negatives may remain, which is a common challenge in computational AMP research. Nevertheless, PLMs help our method remain robust to such noise⁵⁰. Assembling a negative dataset validated by experiments would be valuable in future studies.

Host (animal-derived) AMPs, key components of innate immunity, have received limited attention and lack high-throughput screening methods, despite their potential for low toxicity and therapeutic use. Our analysis showed animal-derived AMPs have unique length distributions and probably undiscovered diversity, highlighting the need for better detection tools. Performance and experimental validation confirm that HMD-AMP effectively addresses this need. Cross-host validation also demonstrates HMD-AMP’s broad applicability: we experimentally confirmed functional AMPs from livestock, wild primates and humans, spanning diverse evolutionary backgrounds. The consistently high success rate in cattle highlights HMD-AMP’s agricultural potential, while slightly lower rates in Hadza and baboon cohorts probably reflect greater sequence divergence in their microbiota^{44,45}.

While integrating metaproteome data ensure peptide expression, protein detection in environmental samples does not guarantee biological functionality in vivo; it can be affected by factors such as stability, host interactions and the local microenvironment^{51,52}. Some peptides may lack antimicrobial activity in living systems or be active only under specific conditions. Our in vitro and in vivo assays confirmed the antimicrobial activity of several AMPs, supporting the biological relevance of metagenome-derived candidates. However, broader functional assays are necessary to fully assess their therapeutic potential.

AMP’s activity in vivo is highly susceptible to proteases in the bloodstream. In our study, we observed discrepancies between in vivo and in vitro activity: Swine_2 exhibited strong antimicrobial activity in vitro but only modestly increased the survival rate of mice to 30%. We performed a literature review about the in vivo experiments conducted in previous AMP studies^{9–14,17} and found that all are within skin and lung infection models. Compared with our mouse model, which easily causes systemic effects, infections in previous experiments are almost localized (Fig. 5e), and our raw peptide is not optimized, which may explain the reduced effect of Swine_2 in vivo.

In the future, our AMP discovery pipeline could be improved by adding specialized modules to identify AMPs suitable for practical use, focusing on low toxicity and strong in vivo stability. Integrating peptide

stability, toxicity and MIC prediction steps would help to further narrow down candidate sequences, making the process more efficient and reducing the time and resources required for synthesis and testing.

Methods

Data used in the study

Different types of datasets were used in our study, including the training set, the test set, the AMP targets prediction data, swine genome data, swine gut microorganism genome data and proteome data.

Training data. The training dataset for this paper preserves the length distribution and quantity balance of positive and negative datasets. The positive dataset consists of 16,916 non-redundant sequences obtained from DBAASP, LAMP2 and DRAMP databases. These peptides were checked for redundancy using the CD-HIT⁵³ program with a threshold of 90%, yielding 5,985 AMPs as the final positive data.

The negative training dataset was generated from the UniProt³² query:

‘cytoplasm NOT antimicrobial NOT antifungal NOT antibiotic NOT antiviral NOT effector NOT excreted’, and results in 574,889 sequences. We then apply AMPgram⁵⁴ to generate non-AMPs and randomly choose each non-AMP on the basis of the length distribution of positive samples. Specifically, we ensure that every AMP has a corresponding non-AMP of the same length. In the end, the negative dataset with 5,985 non-AMPs was generated. The length distribution of both positive and negative datasets is identical.

Test data. Sidorczuk et al.⁵⁵ found training and benchmark datasets produced by a similar negative data sampling method could affect model performance positively and thus cannot reflect the authentic performance of the model. To alleviate the case, we curated four sub-datasets using different pipelines. Subset 1: We collected AMPs used in the work of Su et al.⁵⁶. Peptides that are 90% redundant with our training data are removed beforehand, leaving a dataset of 666 peptides. Subset 2: positive datasets that were implemented by Ma et al.⁹ to test their ensemble models. The dataset is compiled via UniProt, ADAM, APD, CAMP and LAMP databases, which have 1,085 AMPs. Subset 3: We sampled 1,085 non-AMPs that are shorter than 50 AAs from the negative test set of Ma et al.⁹. Most of the non-AMPs are longer than 30 AAs. Subset 4: We randomly sampled 3,118 non-AMPs from the work of Sidorczuk et al.⁵⁵, with most of them being shorter than 30 AAs. After removing the duplicate sequences and identical sequences with training data, the test set consists of 1,234 positive samples and 4,121 negative samples.

AMP targets prediction data. Data used for AMP target prediction are collected from two resources: the DBAASP database and the AMPfun study. We extract the data for each target separately. Taking the Gram-negative type as an example, for positive data, we first combined AMPs against Gram-negative bacteria in DBAASP and AMPs targeting Gram-negative bacteria in AMPfun. Then, CD-HIT is applied to filter positive sequences by a similarity threshold of 60%, and the final positive data are obtained. For negative data, we directly adopted the negative data provided in AMPfun. We have six datasets corresponding to six different AMP functional types for training and testing.

Genome of unknown species from nine mammalian origins. We collected the genomes of nine mammalian origins’ gut microorganism from the National Center for Biotechnology Information (NCBI). Then, taxonomic annotation of each genome was performed using GTDB-Tk v2.1.1 with the release207_v2 databases⁵⁷. Genomes that lacked GTDB annotation at the species level were identified as unknown species. Here, around 200 unknown species, each with one representative genome with high (completion >90% and contamination <5%) and medium quality (completion >50% and contamination <10%), were selected per mammalian for the prediction of sORFs.

Pig genome and gut microbial genomes. We gathered all prokaryotic isolate genomes and metagenome-assembled genomes (MAGs) from the swine gut microbiome (publicly available as of March 2022). We compiled the isolate genomes from the pig intestinal bacterial collection (PIBAC, PRJNA561470), as well as cultured swine gut genomes available in the NCBI and Global Microbial Gene Catalog. In addition, we included all of the gut MAGs generated in Chen et al. (CNP0000824), Youngblut et al. (PRJEB38078) and Zhou et al. (PRJNA647157). To standardize the genome quality across all sets, only high- (completion >90% and contamination <5%) and medium-quality (completion >50% and contamination <10%) genomes were kept for the prediction of sORF. The final number of genomes matching these criteria resulted in a total of 14,026 pig gut microbial genomes.

The annotated reference genome from a Duroc pig (Sscrofa11.1, GenBank accession no. [GCF_000003025.6](https://www.ncbi.nlm.nih.gov/GenBank/000003025.6)) was downloaded from the NCBI and used for the mining of the host AMPs.

HMD-AMP framework

HMD-AMP takes the raw AA sequence as input and deploys a two-level prediction strategy that first predicts whether the sequence is an AMP. If it is predicted as an AMP, our model then annotates its functional types. The architecture of HMD-AMP comprises two key components: a feature extraction module to generate embeddings of peptides and a prediction module to recognize AMPs and their function types.

Feature extraction module. We take the 650M version of ESM-2 as the underpinning of the feature extraction module, which can generate a 1,280-dimensional vector for every residue. To get all the input sequences’ protein-level embeddings with equal length as the inputs to the prediction module, we average across all residue positions of the residue-level embeddings, hence getting a 1,280-dimension feature vector for each protein sequence.

As ESM-2 is not dedicated to short peptides, to facilitate our task, we finetune the ESM-2 on our training data. Specifically, we build a multilayer perceptron (MLP) on top of ESM-2; the MLP is mainly composed of three linear layers and the detailed information is presented in Supplementary Table 1. We subsequently freeze all the layers in ESM-2, and only the parameters in MLP are trainable. For the AMP/non-AMP prediction task, the training data are balanced, and we use the binary cross entropy as the loss function, which can be written as:

$$\mathcal{L}_{\text{BCE}}(y, \hat{y}) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})), \quad (1)$$

where y represents the true binary label, and $\hat{y}$ represents the predicted probability of the positive class. For the AMP functional activities prediction task, the training data for some types (cancer, mammalian cell and virus) are extraordinarily imbalanced. We used the label-distribution-aware margin (LDAM) loss function⁵⁸. The LDAM loss aims to enhance the cross-entropy function by incorporating a margin term Δy , thereby improving the generalization of different classes. It suggests that an appropriate margin should effectively strike a balance between the generalization of major classes and minor classes. The class-dependent margin for our task can be written as:

$$\Delta y_j = \frac{C}{n_j^{1/4}} \text{ for } j \in \{0, 1\}, \quad (2)$$

where n_j is the sample size of the positive or negative class, C is a hyper-parameter that can be tuned and we adopt $C = \frac{0.5}{\max_{0 \leq j \leq 1} (n_j^{-1/4})}$. The LDAM loss can be written as

$$L_{\text{LDAM}}(x, y) = -\log \left(\frac{e^{(z_y - \Delta y)}}{e^{(z_y - \Delta y)} + \sum_{i \neq y} e^{z_i}} \right), \quad (3)$$

where z_y denotes the logit score of the ground truth label and z_i denotes the logit score of the opposite label.

We extract the 320-dimensional vectors from the output of the penultimate linear layer of the fine-tuned PLM as the embeddings of peptide sequences. The embeddings are then fed into the prediction module as the inputs.

Prediction module. Deep forest²⁷ serves as the classifier to identify AMPs and predict their functional types. Deep forest demonstrates a cascade forest structure, and each level of the cascade receives the feature information processes from its previous level and feeds its outputs to the next level as inputs. Here, in each level, our prediction module has four estimators, and a single estimator is an Extreme Gradient Boosting (XGBoost)⁵⁹ with 1,000 trees for learning by minimizing the following objective function:

$$\mathcal{L}(\phi) = \sum_i l_{\text{BCE}}(\hat{y}_i, y_i) + \sum_k \Omega(f_k), \quad (4)$$

where

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w_k\|^2. \quad (5)$$

Here, $l_{\text{BCE}}(\hat{y}_i, y_i)$ is the binary cross-entropy loss of predicted output $\hat{y}_i$ of the model and the ground truth label y_i , and $\Omega(f)$ penalizes the complexity of the model to smooth the final weights and avoid overfitting. T is the number of trees in the ensemble, γ is the complexity regularization parameter, λ is the L2 regularization parameter and w_k are the weights of the individual trees. The AMP/non-AMP prediction is a binary classification task, so further data processing is not needed. However, the AMP target prediction is a multilabel classification task, and we convert this problem into a binary classification by curating the dataset for each target and training it separately.

Training details. In the fine-tuning stage of PLM, we apply the AdamW optimizer⁶⁰ with the initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-3} . We set the number of epochs to 50, with the early stopping strategy to get the best model during training. For AMP target prediction, LDAM loss with class-balanced reweighting is applied and we assign weights for positive and negative classes on the basis of the inverse of sample frequencies.

The above hyperparameters are selected artificially through multiple rounds of fivefold cross-validation. Then, at the training stage, after the hyperparameters are determined, the complete training set is used for training in the AMP/non-AMP prediction task. For the AMP target prediction task, the hyperparameters are inherited and the fivefold cross-validation strategy is used for training and evaluation: In each iteration, one of the folds is used as the test set while the remaining four folds are used for training. This process is repeated five times, with each fold taking a turn as the test set. The final results are the average performance obtained from these iterations.

The fine-tuning, training and performance evaluation of HMD-AMP and other included models are running on two Nvidia GeForce RTX 3090 GPU cards with 24 GB of memory. For other models, we adopted the hyperparameters they provided in the paper to train their model.

Evaluation metrics

Model performance evaluation. When testing the performance of HMD-AMP and other models in the AMP/non-AMP prediction task and AMP target prediction task, we used the area under the receiver operating characteristic curve (presents the true positive rate of a classifier versus its false positive rate at various decision threshold levels), area under the precision–recall curve (to better illustrate the precision–recall trade-off of a model) and the following evaluation metrics to comprehensively analyse the performance:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}, \quad (6)$$

$$\text{Balanced accuracy} = \frac{\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}}}{2}, \quad (7)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (8)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (9)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{recall}}{\text{Precision} + \text{recall}}, \quad (10)$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}}, \quad (11)$$

where TP denotes the number of true positives, TN denotes the number of true negatives, FP denotes the number of false positives and FN denotes the number of false negatives.

Similarity estimation. To measure the sequence similarity of peptides, we adopted the Needleman–Wunsch algorithm provided by the ‘pairwise2’ function from the Biopython library. This function optimizes the algorithm so that it can be better parallelized for higher efficiency. A threshold of 40% is commonly used to divide sequences that have high or low similarity with each other⁶¹. To measure the structure similarity of peptides, we used the TM-scores calculated by TM-align³³. TM-score is independent of the size of the protein; it quantifies the similarity between two structures using a score ranging from 0 to 1, where a score of 1 signifies a perfect match between the structures. For the TM-score less than 0.5, proteins are mostly not in the same fold⁶². To create the logo plot, we used WebLogo⁶³. Details of multiple sequence alignment of peptide sequences were visualized using MEGA11 software⁶⁴.

Structure prediction. Considering the computational resource, we applied the ColabFold⁶⁵ version of AlphaFold2⁶⁶ for the tertiary structure prediction of peptides. Specifically, we set the number of synchronously running models as 1, and used the amber relaxation to refine the prediction. Other settings remain the default. For the visualization of the structure and the following alignment operation, we utilized PyMOL⁶⁷.

AMP mining pipeline

sORF prediction. The ‘getorf’ function in the EMBOSS software package (version 6.6.0.0) was used to predict sORFs from our collected gut microbial genomes. The parameters were set as ‘find 2 -table 11 -minsize 15 -maxsize 300’. For the swine host genome, annotated protein sequences from a Duroc pig (Sscrofa11.1, GenBank accession no. GCF_000003025.6) were directly downloaded from the NCBI and filtered for short proteins ($5 < \text{length} < 100$). Then, the sORF dataset from microbial genomes and the host genome was fed into our prediction model to identify potential AMPs.

Filter potential AMPs with metaproteome. To ensure that microbial sORFs predicted in silico are probably expressed to proteins/peptides, two metaproteome datasets (PXD006224 and PXD027185) from ProteomeXchange⁶⁸ were collected and used to select AMP sequences. The raw mass (tandem mass spectrometry) spectra data were converted into Mascot Generic Format (mgf) files by pParse and screened against the sORFs database using an iterative search algorithm with DBReducer.

Motif discovery for remote AMPs. Motif discovery was conducted using STREME⁴⁶. Both the AMP sequences from the training dataset and the 48 discovered remote AMP sequences were used as input, while

non-AMP sequences from the training set served as the control. Motif widths were constrained to 8–30 AAs. STREME evaluates motif enrichment using a statistical test with a significance threshold of $P < 0.05$, which also serves as the stopping criterion.

Peptide synthesis

The peptides used in this study were synthesized by Bankpeptide Biological Technology using solid-phase peptide synthesis. Their molecular weights were confirmed by mass spectrometry (molecular weight confirmation), and high-performance liquid chromatography analysis verified a purity of over 95% for all compounds.

Bacterial inhibition experiment

Bacteria used for validation assay include *S. aureus* (ATCC 6538), *E. coli* (ATCC 25922), *S. enteritidis* (China Center for Type Culture Collection (CCTCC) 20082847), *S. suis* (National Center for Veterinary Culture Collection (CVCC) 4268) and ETEC K88 and ETEC K99 (donated by the China Agricultural University). Target bacteria were streaked on Mueller–Hinton agar medium and incubated at 37 °C overnight. The individual colonies were picked into MHB medium and shaken at 220 rpm at 37 °C overnight. The MHB bacterial solution was diluted to the initial concentration (OD_{600} of 0.1) and then diluted again 200 times for the bacterial inhibition test. The freeze-dried powder of AMPs was dissolved in double-distilled water to a concentration of 2.4 mmol l^{-1} . The bacterial inhibition test of AMP contains four experimental groups: (1) blank control group A, 100 μl of MHB solution; (2) blank control group B, 97.5 μl of MHB solution and 2.5 μl of AMP solution; (3) bacterial control group, 97.5 μl of bacterial solution and 2.5 μl of double-distilled water; and (4) AMPs test group, 97.5 μl of bacterial solution and 2.5 μl of AMP solution, with a final concentration of AMPs of $60 \text{ }\mu\text{mol l}^{-1}$. Experiments were performed on 96-well plates with each single well containing 100 μl of final volume. The OD_{600} value of each well was measured after culturing at 37 °C for 24 h. The normalized OD_{600} was calculated as follows:

$$\frac{OD_{\text{AMPs test group}} - OD_{\text{blank control group B}}}{OD_{\text{bacterial control group}} - OD_{\text{blank control group A}}}, \quad (12)$$

and all experiments were performed with three independent replicates.

MIC assay of AMPs

MIC determination of AMPs was performed by broth microdilution. Pathogens were inoculated in MHB medium and shaken at 220 rpm at 37 °C overnight. The cultures were diluted to the initial concentration (OD_{600} of 0.1) roughly equivalent to 1×10^8 colony-forming units (CFU) ml^{-1} and then again diluted 200 times for the MIC test. Each MIC test group contains ten levels of concentration of AMPs by twofold serial dilution, from $64 \text{ }\mu\text{mol l}^{-1}$ to $0.125 \text{ }\mu\text{mol l}^{-1}$. A total of 100 μl bacterial solutions containing different levels of AMPs were transferred into 96-well plates. MIC assay also sets a negative control group containing 100 μl MHB solution and a positive control group involving 100 μl bacterial solution. After incubating at 37 °C for 24 h, the OD_{600} was measured. The MIC was determined as the lowest level of AMPs with an OD_{600} value within 3 s.d. from the mean of the negative control. All assays were performed with three independent replicates.

Haemolysis effect assay

The supernatant was removed from the 4% swine red blood cell solution (Sbjbio) after centrifugation at 376g for 5 min and then resuspended with PBS buffer containing different concentrations (128, 64, 32, 16, 8, 4, 2, 1, 0.5 and $0.25 \text{ }\mu\text{mol l}^{-1}$) of AMPs. The mixtures were centrifuged at 845g for 10 min after 1 h reaction at 37 °C. The supernatant was separated and the absorbance was measured at 570 nm. All experiments were performed with three independent replicates.

Cytotoxicity against mammalian cells test

Growing HT-29 cells (purchased from ATCC) were seeded in 96-well cell culture plates in Dulbecco's Modified Eagle Medium basic (DMEM, 1 $\times$; Gibco) with 10% (vol/vol) fetal bovine serum (Gibco) and 1% (vol/vol) penicillin–streptomycin liquid (Solarbio), after 12 h incubation at 37 °C with 5% CO_2 in the atmosphere. Then, the medium was replaced with DMEM basic culture medium with 10% (vol/vol) fetal bovine serum and various concentrations (128, 64, 32, 16, 8, 4, 2, 1, 0.5 and $0.25 \text{ }\mu\text{mol l}^{-1}$) of AMPs, followed by 24 h incubation. Cell viability was monitored using an MTT Cell Proliferation and Cytotoxicity Assay Kit (Beyotime), and the absorbance was measured at 570 nm. Each AMP experiment was performed with four independent replicates.

SEM measurement

ETEC K88 cells were washed and resuspended with PBS buffer at an OD_{600} of 0.3 and then incubated with AMP solution (at 4 $\times$ the MIC final concentration) at 37 °C for 3 h. Subsequently, the bacterial cells were collected by centrifugation at 6010g for 3 min, fixed with 0.5 ml 2.5% (vol/vol) glutaraldehyde and placed at 4 °C for 24 h. Hangzhou Yanqu Information Technology performed the following sample pretreatment steps and SEM measurements. In brief, the fixative solution was washed three times with PBS buffer for 15 min each time, and then dehydrated through a graded ethanol series (30%, 50%, 70%, 80%, 90% and 95%, with each step lasting 15 min) before being placed in 100% ethanol for 20 min. After dehydration, the samples were placed into a critical point dryer. The dried samples were fixed on the sample stage with conductive carbon glue and sprayed with Pt using an ion-type sputtering instrument for 2 min. The treated samples were observed using a scanning electron microscope (Hitachi Regius 8100).

Animal experiments

We eventually selected one peptide to confirm its effectiveness against ETEC K88 (donated by China Agricultural University, Beijing, China) in a murine acute peritonitis mouse model. First, a single colony of ETEC K88 was incubated in the MHB medium and grown to the exponential phase, bacterial cells were collected and the concentration controlled to $5 \times 10^8 \text{ CFU ml}^{-1}$ with PBS buffer. Then, a murine acute peritonitis model was performed on female BALB/c mice (6 weeks) with 0.2 ml above-mentioned bacterial solution. Specifically, all mice were kept in the same temperature and humidity room on a 12-h light–dark cycle and were fasted for 12 h before infection. They were given an intraperitoneal injection of 0.2 ml sterile saline (control group), 0.2 ml sterile saline with 10 mg kg^{-1} (mg per kg body weight) AMP or 0.2 ml sterile saline with 10 mg kg^{-1} gentamicin 0.5 h after administration of the bacterial solution, respectively. The survival rate was monitored for 7 days. All animal experiments were approved by the Ethics Committee of the Shenzhen Institute of Advanced Technology, Chinese Academy of Science (Institutional Animal Care and Use Committee (IACUC) no. SIAT-IACUC-240513-HCS-DL-A2818).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All the datasets we used are provided in the Methods and are publicly available. The training dataset for HMD-AMP and predicted sORFs of all nine mammals is available via Zenodo at <https://doi.org/10.5281/zenodo.15622525> (ref. 69). All other relevant data supporting the key findings of this study, such as the results in the AMP discovery, are available within this Article and its Supplementary Information. Source data are provided with this paper.

Code availability

The open source codes of HMD-AMP, including trained weights, inference scripts, and training scripts, are available via GitHub at

<https://github.com/ml4bio/HMD-AMP>. Logo plots were created in WebLogo (<https://weblogo.threeplusone.com/>). Visualization of multiple sequence alignment of peptides used MEGA11. Structure visualizations used PyMOL (<https://pymol.org/>). Helical wheel plots used HeliQuest (<https://heliquest.ipmc.cnrs.fr>).

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Acknowledgements

This study was supported by the Shenzhen Medical Research Fund (award no. A2503002 to Y.L. and B2302036 to L.D.), the National Key R&D Program of China (grant no.2025YFA0923500 to Y.L.), the Chinese University of Hong Kong (CUHK; award nos. 4937025, 4937026, 5501517, 5501329 and SHIAE BME-p1-24 to Y.L.), the IdeaBooster Fund (award nos. IDBF23ENG05 and IDBF24ENG06 to Y.L.) and the Shenzhen Key Laboratory for the Intelligent Microbial Manufacturing of Medicines (award no. ZDSYS20210623091810032 to L.D.) and partially supported by grants from the Research Grants Council of the Hong Kong Special Administrative Region (Hong Kong SAR), China (project nos. CUHK 24204023 and 14208525 to Y.L.), and grants from the Innovation and Technology Commission of the Hong Kong SAR, China (project nos. GHP/O65/21SZ, ITS/247/23FP and PRP/O33/24FX to Y.L.). This research was also supported by the Research Matching Grant Scheme at CUHK (award nos. 8601603 and 8601663 to Y.L.) and the National Natural Science Foundation of China (award no. 32201313 to H.L.). We thank P. Wang and Q. Shen from the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, China, for their help with animal experiments. We also thank S. Qiao from the China Agricultural University, China, for providing the enterotoxigenic *Escherichia coli* K88 and K99 strains used in this work.

Author contributions

Q.Y., H.L., H.S., Z.D., L.D. and Y.L. conceived the research; Q.Y., H.L. and Y.A. collected the dataset; Q.Y. implemented the main algorithm; Q.Y., H.L., H.S., J.S., C.Z. and L.S. performed the experiments; Q.Y., H.L., H.S., J.S. and L.Z. conducted the analysis; Q.Y., H.L. and H.S. wrote the paper; and L.D. and Y.L. supervised the project. All authors read and approved the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-026-01630-w>.

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Peer review information *Nature Biomedical Engineering* thanks Martin Steinegger and Dong Xu for their contribution to the peer review of this work. Peer reviewer reports are available.

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datasets (<https://github.com/zhanglabNKU/APIN>, <https://github.com/BioGenies/NegativeDatasets>, https://github.com/mayuefine/c_AMPs-prediction, and <http://fdlab.csie.ncu.edu.tw/AMPfun/index.html>).

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Cell line source(s)	HT29 cells (female origin) were purchased from ATCC.
Authentication	Cells were authenticated by the vendor and no further authentication was performed in the laboratory.
Mycoplasma contamination	There is no Mycoplasma contamination in cell experiments.
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Laboratory animals	Animal tests performed on female BALB/c mice (6 week-old), all mice were kept in the same temperature and humidity room on a
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Laboratory animals	12-hour light/dark cycle and were fasted for 12 hours before infection. The used bacteria, Enterotoxigenic Escherichia coli (ETEC) K88, was donated by China Agricultural University, Beijing, China. Source of other strains: Staphylococcus aureus ATCC 6538 and Escherichia coli ATCC 25922 were purchased from the American Type Culture Collection (ATCC; Rockville, MD). Salmonella enteritidis CCTCC 20082847 was purchased from the China Center for Type Culture Collection. Streptococcus suis CVCC 4268 was purchased from the National Center for Veterinary Culture Collection. ETEC K99 was donated by China Agricultural University, Beijing, China.
Wild animals	NA
Field-collected samples	NA
Ethics oversight	All animal experiments were approved by the Ethics Committee of the Shenzhen Institute of Advanced Technology, Chinese Academy of science(IACUC number: SIAT-IACUC-240513-HCS-DL-A2818)

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